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Bachelor Project Writing Guide

media/image21.png ***Panduan Penulisan Projek Sarjana Muda***

[**version 2025**]

JAWATANKUASA PSM & PD 2025

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CHAPTER 1. INTRODUCTION Introduction

Introduction to your project and your project background as a whole but in brief.

Problem statement(s)

- Description of problems that directly influence the motives of the project. (Example, explanation of problems of XYZ & Co. record keeping)
- Applicable for project of improving or solving a specific case.
- Project that is of new creation, re-creation, individual initiative or not specifically involves any case to deal with, the problems can be included in the preamble paragraphs.

Objective

Can be in point form format (use bullets) together with a brief explanation.

Scope

- Describes every scope involved in your project and give reason(s) for the involvement. Examples: (e.g. children education), specific users (e.g. children between 5 and 8), other specific entities, specific platform (e.g. network, OS) etc.
- Can be in point form format (use bullets) together with a brief explanation.

Project Significance

Describes who/what may benefits from the project and how? You can infer it back to your objectives (if applicable).

Expected Output

Describes what do you expected from your project.

Conclusion

Give a summary of this chapter and next activities to be developed.

CHAPTER 2. LITERATURE REVIEW AND PROJECT METHODOLOGY Introduction

Preview to the literature review and project methodology.

Facts and findings (based on topic)

1. **Domain**

   Identify domain related with your project with explanations.

Existing System

- Identify domain related with your project with explanations.
- Discuss and state your approach and related or past research, references, case study and other finding that relate to your project title. Examples: readings 1 and 3, experiments you get 3 and 4, case study 5 and 6 etc.
- Tagged the source if you refer the approach from published materials.
- Support the approach with statements (from published materials) to justify you fact findings are sound.
- Identify hardware and software used

Technique

- State other approaches than what you use that you think also applicable and related and give reason to justify why not.

Project Methodology

- Describe the selected approach or methodology used in your project (Examples: SSADM/SDLC/OOAD, Instructional design, Database life cycle and relevant methodology)
- Describe the activities that you may do in every stage and relate with your project.
- Include statements (from published materials) that support the approach you apply.

Project Requirements

1. **Software Requirement**
    - List software requirements in point form. Examples: MS Visual Ba- sic Professional v.6.x, Apache Tomcat and the related for developing application software, MS Project 2000 for project management etc.

Hardware Requirement

- List hardware requirements in point form. Examples: PCs, server, devices and storage.

Other Requirements

- State other requirements to be used in your project such as you need special lab for project development, photo copy facility, discussion room and etc.

Project Schedule and Milestones

- Explain your actions plan prior to the end of the project. Apply from what you have learnt from project management.
- List and describe stage by stage of your activities.
- Attach your project time line or Gantt chart (in 1 page view)

Conclusion

Summarize the chapter and explain the next activities to be developed.

CHAPTER 3. ANALYSIS Introduction

Preview to the analysis phase and how it would be developed.

Problem Analysis

- Investigate and describe current system scenario/situation. Reiterate the data flow diagram or activity diagram from your reference(s), showing how the current system(s) or business(s) runs.
- Use appropriate diagram to visualize the system flow such as sequence diagram (if use OOAD with UML), DFD (if use SSADM) and Flowchart (for network project).
- For simulation based project, analyse the real graph given by company.
- Explain in detail the problems statement as mentioned in Chapter 1.

Requirement analysis

1. **Data Requirement**
    - What data should the system input and output, and what data should the system store internally.
    - It can be illustrated by using Data Model or Data Dictionary.

Functional Requirement

- Specify the functions of the system, how it records, compute, transforms, and transmits data.
- It can be illustrated by using DFD, Context diagram or Use case.

Non-functional Requirement

Specify how well the system performs its intended functions. E.g. Quality requirement, Performance-how many computer resources should it use, how accurate should the result be, how much data should it be able to store?

Others Requirement

Describe each of software, hardware and requirements that will be used (justification of usage).

Conclusion

Summarize the chapter and explain the next activities to be developed.

CHAPTER 4. DESIGN Introduction

Introductory preview to this chapter. Examples: this chapter defines the results of the analysis of the preliminary design and the result of the detailed design.

High-Level Design

Describe high-level view of your system's structure or system's interior. You may refine some of the function below once you are in PSM II.

System Architecture

- Define architecture view of your system. The architecture view can be presented in layer, framework, tier or patent (for OOAD).
- Describe the static view and dynamic view of your application (for OOAD - use interaction diagram, high-level class diagram).
- For those use SSADM: use any appropriate diagram to view your system.

User Interface Design

Refine your user interface design that defined in chapter 5 of PSM I.

a. Navigation Design

Define and refine the navigation flow and types of navigation controls.

b. Input Design

- Define and refine the screens (e.g. types of inputs such as text, numbers, and selection box etc.) used to enter information, as well as any forms on which users write or type information.
- Define and refine validation rule for each of input field.

c. Output Design

Define and refine the types of outputs including detail reports, summary reports, turnaround documents and graphs. Classify your output in term of periodically or ad-hoc basis. For example daily, monthly, yearly etc.

Database Design

1. **Conceptual and Logical Database Design**

    - Introduce briefly to the logical data model (LDM) or entity relationship diagram (ERD) and what they have to do with your database design.
    - Define, refine and construct the entity relationship (ER) diagrams in details with explanation in text on what basis (business rules) you apply for every entity relationship you have defined in the diagram.
    - Data dictionary and normalization

Detailed Design

Specification and diagrams may be further elaborated. Emphasis should be on the logic of the design and the approach to satisfying the requirements – How will the system function?

Software Design

- For those use SSADM/SDLC: Describe in detail of every functions according to your DFD in format of program specification. The program specification includes information of program description, file input/output, pseudo code and attach sample screens.

- For those use OOAD/UML: Describe in detail of every classes. Examples: class name, responsibility, attributes methods/operations. Describe every methods/operations such as its responsibility, input/output parameter, pre/post condition and algorithm –should be bias to language chosen for your software development (e.g. in English like of Java code).

Physical Database Design